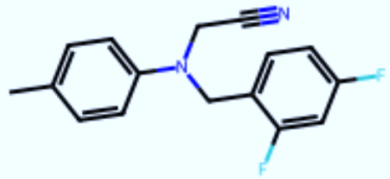
| | Smiles | Standard Type | Standard Relation | Standard Value | Standard Units | Assay ChEMBL ID | ROMol |
|---|---|---|---|---|---|---|---|
| | | | | | | EMBL3625335 |  |
| | @H]1CC[C@@H]2CN1C(=O)N2OS(=O)(=O)[O-].[Na+] | MIC | '<=' | 0.120 | ug.mL-1 | CHEMBL5247824 |  |

# DSA103 Advanced Chemical Data Science

Lecture 12: Life Science Data – Data Structures / Molecular Representations

Dr. Johannes Schörgenhumer (johannes.schoergenhumer@chem.uzh.ch)

# Motivation

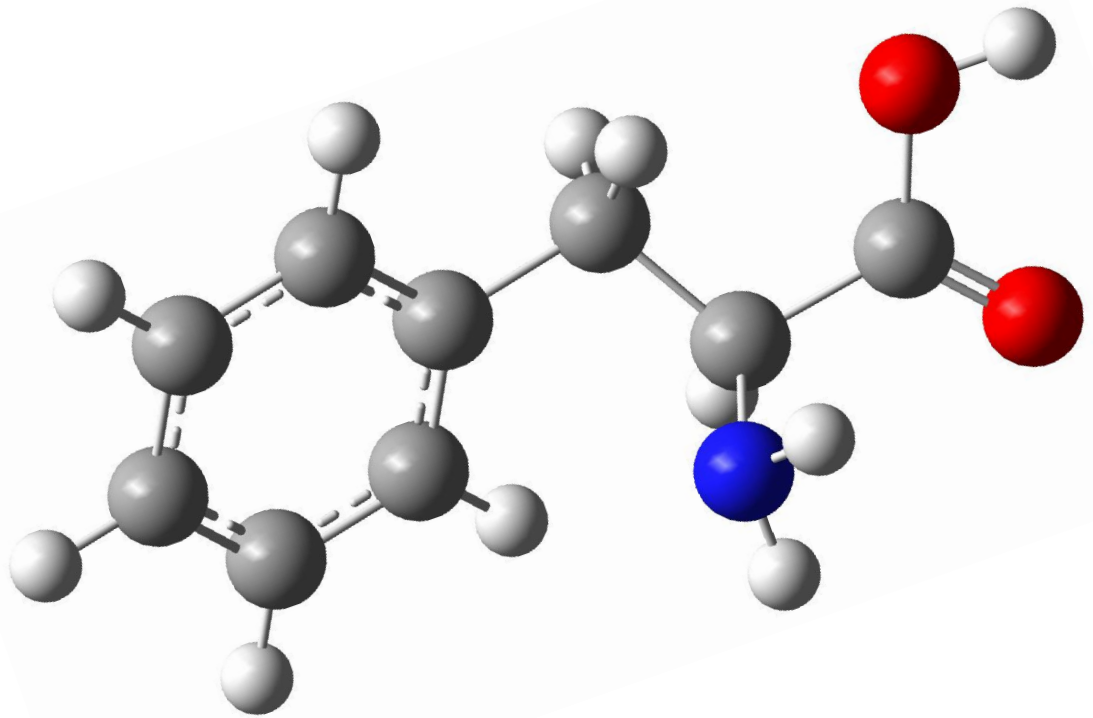
Data Structures: How data types are actually encoded, as strings, matrixes, coordinate lists, ...

Today we look again at molecular representations!

Representations translate chemistry into data!

Different means (e.g. 1D, 2D, 3D, fingerprints, reactions)…

…for different tasks (e.g. property prediction, structure search)



# Content overview

Chemical Notations:

- SMILES
- Bit vectors / Fingerprints

Mathematical Notations:

- Graphs
- Matrices
- Connection tables

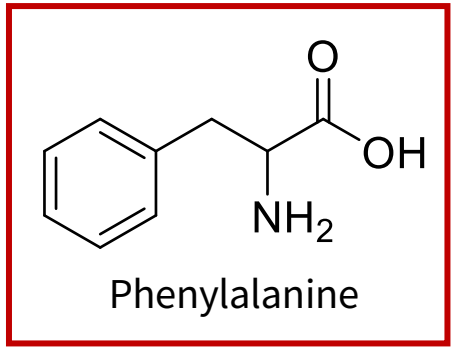
3D representations and some data formats

Encoding chemical reactions

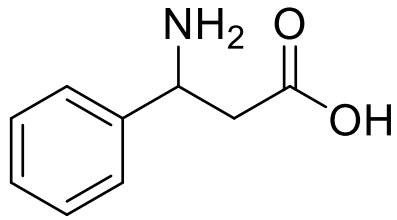
# Encoding molecular structures – SMILES:

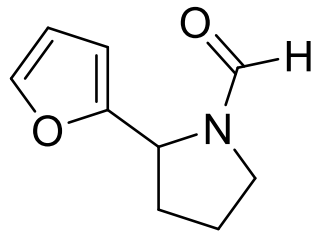
1D chemical notations:

**Empirical formula** / “Sum formula”: e.g. $C_9H_{11}NO_2$



Phenylalanine




**SMILES**: NC(Cc1ccccc1)C(O)=O
Or: C1=CC=C(C=C1)C[C@@H](C(=O)O)N
Or: NC(C(=O)O)(CC1=CC=CC=C1)

**SLN** (SYBYL Line Notation): C[1]H:CH:CH:CH:CH:C(:@1)CH2CH(NH2)C(=O)OH

**InChI** (IUPAC International Chemical Identifier): 1S/C9H11NO2/c10-8(9(11)12)6-7-4-2-1-3-5-7/h1-5,8H,6,10H2,(H,11,12)/t8-/m0/s1

- Normalized (aromaticity, tautomers)
- Canonicalized (unambigous labelling)
- Serialized (linear string as output)

Chemical formula | Connection sublayer | H atom sublayer | stereochemistry

**InChIKey:** Compressed, not human-interpretable 27 character string computed from InChI for facilitated search, e.g. Phenylalanine = COLNVLDHVKWLRT-QMMMGPOBSA-N

Engel T. (2018) *Principles of Molecular Representations* in Engel, T., Gasteiger, J., *Cheminformatics* (1st ed., pp. 9-41), Wiley-VCH.

# Encoding molecular structures – SMARTS: SMILES as a query language

Every atom and bond becomes a pattern to search for substructures, not whole molecules

| Pattern | Matches |
|---|---|
| `[#6]` | any carbon, aliphatic or aromatic |
| `C / c` | aliphatic / aromatic carbon only |
| `[NX3;H1,H2]` | N with 3 connections and 1 or 2 H |
| `[C,N;!R]` | C or N (comma = OR), not in a ring |
| `* and ~` | any atom and any bond |
| `C@C` | ring bond between two aliphatic C |

```
patt = Chem.MolFromSmarts(
   '[CX3](=O)[OX2H1]')
mol.HasSubstructMatch(patt)
mol.GetSubstructMatches(patt)
```

**Careful**

Any SMILES is valid SMARTS, but the meaning shifts: C now matches in SMILES is an aliphatic carbon, whatever its hydrogens. Always parse queries with MolFromSmarts, never MolFromSmiles.

https://www.daylight.com/dayhtml/doc/theory/theory.smarts.html

# Acitivity: SMARTS Exercises

1) Quick demo
2) Follow the notebook smarts_exercises.ipynb

# Encoding molecular structures – Bit Vectors / Fingerprints:

Bit vectors: “Barcode” for molecular structure:

- Boolean array, sequence of bits (e.g. based on structural features)
- Linear binary strings of fixed length
- Fast to compute
- “One-way” encoding

As bit vectors, structure can be encoded into:

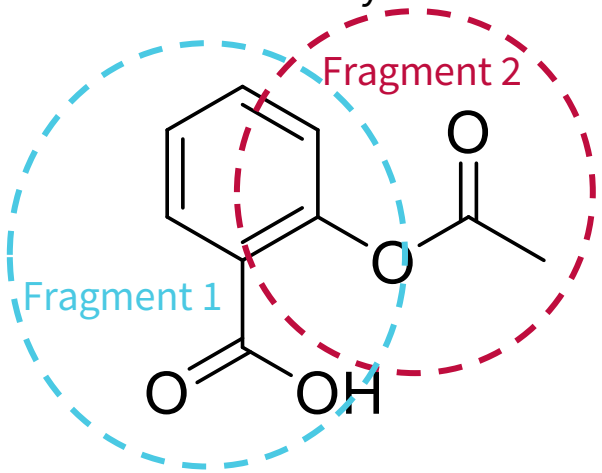
- Hash codes: typically 64-bit vectors, easily indexed, for storing data and searching (full structures)
- **Structure keys:** Boolean vector, encoding structural features based on predefined fragment diary (“contains N”, “contains rings”, …), e.g. MACCS
- **Fingerprints:** No fragment dictionaries. Algorithm generates a series of patterns (atoms / bond sequences), and generates bit vectors for them (e.g. Morgan)

Fingerprints / Structure keys can be used as descriptors in substructure searches, clustering, similarity assessment (e.g. **Tanimoto similarity**)

Tomczak J. (2018) *Data Types* In Engel, T., Gasteiger, J., *Cheminformatics* (1st ed., pp. 156-168), Wiley-VCH.

# Feature engineering: fingerprints

Encode structure as binary vector!




| | Amine | Aromatic ring | Carbonyl | Hydroxy |
|---|---|---|---|---|
| **Fragment 1** | 0 | 1 | 1 | 1 |
| **Fragment 2** | 0 | 1 | 1 | 0 |

**Extended-Connectivity Fingerprints** (ECFP / Morgan fingerprints):

(Order of) Columns are standardizes in different fingerprint models (like Morgan)

— Local neighborhoods hashed into bit vectors

— Typically: 1024 or 2048 bits

— Well-suited for capturing structural information

— Widely applied in QSAR studies

Fingerprint generators in RDKit:
— **AtomPairFP** = rdkit.Chem.rdFingerprintGenerator.FPType.AtomPairFP
— **MorganFP** = rdkit.Chem.rdFingerprintGenerator.FPType.MorganFP
— **RDKitFP** = rdkit.Chem.rdFingerprintGenerator.FPType.RDKitFP
— **TopologicalTorsionFP** = rdkit.Chem.rdFingerprintGenerator.FPType.TopologicalTorsionFP

https://www.rdkit.org/docs/source/rdkit.Chem.rdFingerprintGenerator.html

# Acitivity: Calculate fingerprints and assess similarity

1) Use the Morgan fingerprint generator in rdkit to create bit vectors for two molecules (SMILES)
2) Assess the similarity of the two molecules using the fingerprint to calculate a **Tanimoto score**
3) Try different fingerprint generators and see how the similarity score changes

Text

Note: There are a few fingerprint generators available in rdkit (see link below):

- Morgan
- Feature Morgan (variation of the Morgan fingerprint)
- RDKit
- Atom pairs
- Topological torsions

https://greglandrum.github.io/rdkit-blog/posts/2023-01-18-fingerprint-generator-tutorial.html

Depend on application, community conventions, used fingerprint
State which cutoff and which fingerprint is used

**@Tanimoto similarity = similarity of binary vectors (not only "how alike are two molecules?")**

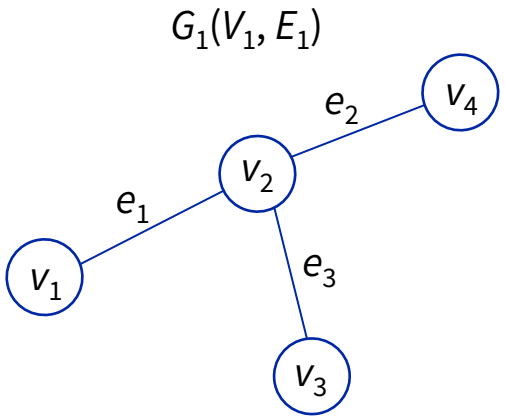
Very dependent on the fingerprint type / binary vector and on the applications:

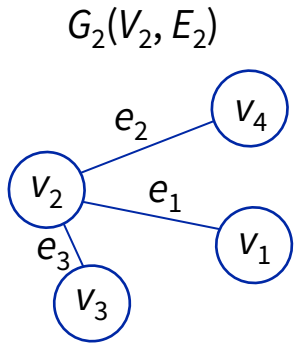
- **≥ 0.85 (Daylight fingerprints):** the classic "similar bioactivity" cutoff. Studies later showed it is far less reliable than its reputation, since a substantial share of pairs above it differ in activity.
- **≈ 0.4 to 0.5 (ECFP4/Morgan r=2):** often cited as the equivalent of 0.85 for Daylight. ECFP scores run lower, so the same cutoff would be much too strict.
- **≥ 0.9 (PubChem fingerprints):** the cutoff PubChem uses for its "similar compounds" neighbours.
- **≈ 0.7 to 0.8 (ECFP4):** usually treated as "close analogs", often single-substituent changes, and used to flag near-duplicates.

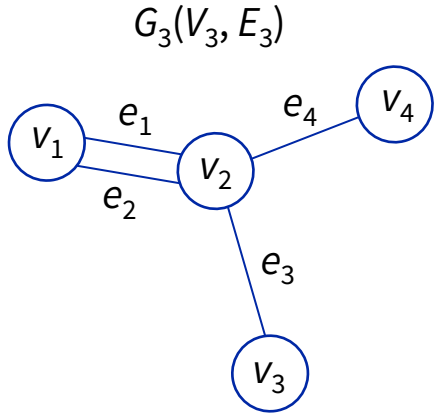
# Mathematical Notations

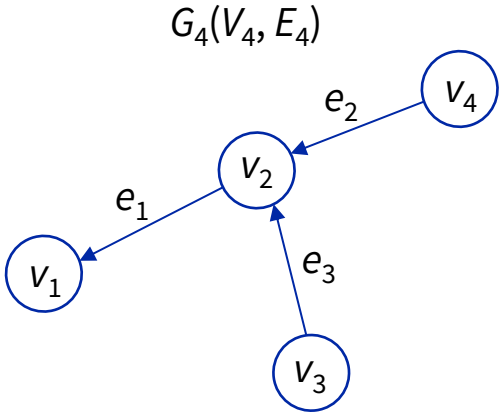
**Graphs:**

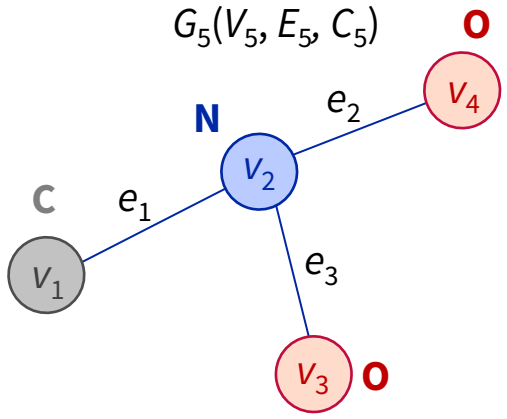
- Structures can be imagined as graphs
- Graph = tuple of vertices & edges: $G = (V, E)$
- "Vertices $v_1$ and $v_1$ are *adjacent* if *incident* in a common edge $e_1$"
- Graphs per se carry no 2D or 3D information ($G_1 = G_2$)
- *Multigraphs* have vertices connected by more than one edge
- *Directed graph* / *digraph*: edges are asymmetric
- Can be coded / coloured: $G = (V, E, C)$




Engel T. (2018) *Principles of Molecular Representation* in Engel, T., Gasteiger, J., *Cheminformatics* (1st ed., pp. 9-42), Wiley-VCH.

# Mathematical Notations

**Matrices:**

- Based on graphs
- Can be used as descriptors
- *Adjacency matrix*: represents the connectivity of all atoms
- *Distance matrix*: reflects bond lengths
- *Incidence matrix*: represents the relationships between atoms
- *Bond matrix*: reflects bonds and bond orders

Boolean: Is it connected to Atom_x (0 or 1)

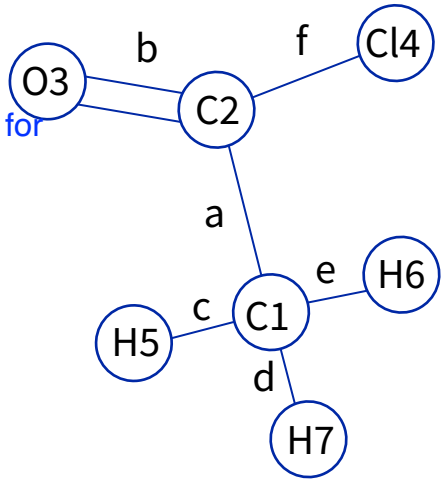
$A_A =$

| | C1 | C2 | O3 | Cl4 | H5 | H6 | H7 |
|---|---|---|---|---|---|---|---|
| C1 | 0 | 1 | 0 | 0 | 1 | 1 | 1 |
| C2 | 1 | 0 | 1 | 1 | 0 | 0 | 0 |
| O3 | 0 | 1 | 0 | 0 | 0 | 0 | 0 |
| Cl4 | 0 | 1 | 0 | 0 | 0 | 0 | 0 |
| H5 | 1 | 0 | 0 | 0 | 0 | 0 | 0 |
| H6 | 1 | 0 | 0 | 0 | 0 | 0 | 0 |
| H7 | 1 | 0 | 0 | 0 | 0 | 0 | 0 |

=

Truncated form to account only for relevant ones

| | C1 | C2 | O3 | Cl4 |
|---|---|---|---|---|
| C1 | 0 | 1 | 0 | 0 |
| C2 | 1 | 0 | 1 | 1 |
| O3 | 0 | 1 | 0 | 0 |
| Cl4 | 0 | 1 | 0 | 0 |




$A_I =$

| | C1 | C2 | O3 | Cl4 | H5 | H6 | H7 |
|---|---|---|---|---|---|---|---|
| a | 1 | 1 | 0 | 0 | 0 | 0 | 0 |
| b | 0 | 1 | 1 | 0 | 0 | 0 | 0 |
| c | 1 | 0 | 0 | 0 | 1 | 0 | 0 |
| d | 1 | 0 | 0 | 0 | 0 | 1 | 0 |
| e | 1 | 0 | 0 | 0 | 0 | 0 | 1 |
| f | 0 | 1 | 0 | 1 | 0 | 0 | 0 |

$A_B =$

| | C1 | C2 | O3 | Cl4 |
|---|---|---|---|---|
| C1 | 0 | 1 | 0 | 0 |
| C2 | 1 | 0 | 2 | 1 |
| O3 | 0 | 2 | 0 | 0 |
| Cl4 | 0 | 1 | 0 | 0 |

Engel T. (2018) *Principles of Molecular Representation* in Engel, T., Gasteiger, J., *Cheminformatics* (1st ed., pp. 9-42), Wiley-VCH.

# Mathematical Notations

Most condensed version on connectivity matrices

**Connection tables:**

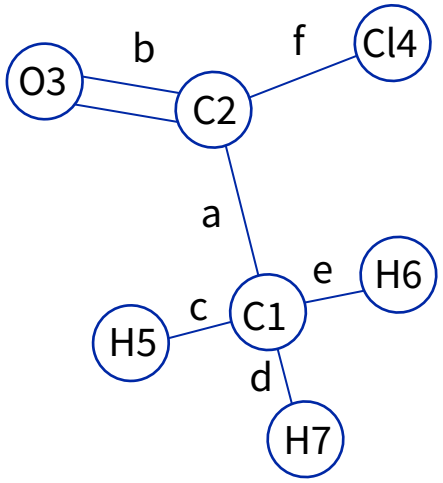
- Based on graphs
- Matrices contain a lot of redundant information
- Atoms are arbitrarily indexed
- Bonds listed with their bond order (BO)
- Atom and bond tables can be combined
- Hydrogens and redundant bond definitions can be omitted

**Atom list**

| Atom index | Element |
|---|---|
| 1 | C |
| 2 | C |
| 3 | O |
| 4 | Cl |
| 5 | H |
| 6 | H |
| 7 | H |

**Bond list**

| 1st atom index | 2nd atom index | bond order |
|---|---|---|
| 1 | 2 | 1 |
| 1 | 5 | 1 |
| 1 | 6 | 1 |
| 1 | 7 | 1 |
| 2 | 3 | 2 |
| 2 | 4 | 1 |




| Atom index | Element | 1st atom index | BO | 2nd atom index | BO | 3rd atom index | BO | 4th atom index | BO |
|---|---|---|---|---|---|---|---|---|---|
| 1 | C | 2 | 1 | 5 | 1 | 6 | 1 | 7 | 1 |
| 2 | C | 1 | 1 | 3 | 2 | 4 | 1 | | |
| 3 | O | 2 | 2 | | | | | | |
| 4 | Cl | 2 | 1 | | | | | | |
| 5 | H | 1 | 1 | | | | | | |
| 6 | H | 1 | 1 | | | | | | |
| 7 | H | 1 | 1 | | | | | | |

Classic format seen in computational data files

| Atom index | Element | 1st atom index | BO | 2nd atom index | BO |
|---|---|---|---|---|---|
| 1 | C | 2 | 1 | | |
| 2 | C | 3 | 2 | 4 | 1 |
| 3 | O | | | | |
| 4 | Cl | | | | |


Engel T. (2018) *Principles of Molecular Representation* in Engel, T., Gasteiger, J., *Cheminformatics* (1st ed., pp. 9-42), Wiley-VCH.

# 3D representations and some data formats

## XYZ format

- Cartesian coordinates for all atoms given
- Constitution and configuration implicitly given
- Similarly encoded in a lot of other formats

Gives real structural information but not precisely, which atoms are connected to which
Often connectivity tables provided together with that

```
1   7
2   acetylchloride
3   C    -6.666666    -1.461295     0.000000
4   C    -6.153324    -0.735339     1.257405
5   O    -6.454696     0.469394     1.460785
6   Cl    -5.141676    -1.593562     2.413961
7   H    -6.310012    -2.470105     0.000000
8   H    -6.309994    -0.956897    -0.873652
9   H    -7.736666    -1.461282     0.000000
10
```

Header with total number of atoms
Optional comment (line not optional!)

Table of cartesian coordinates. No Connection Table!

## MOPAC format

- Z-matrix
- Basically, polar coordinates
- Constitution and configuration implicitly given
- Mostly used in QM and DFT suites (*internal coordinates*)

```
1
2  acetylchloride.mop
3
4     C    0.000000  0    0.000000  0    0.000000  0    0    0    0
5     C    1.540000  1    0.000000  0    0.000000  0    1    0    0
6     O    1.258400  1  120.226946  1    0.000000  0    2    1    0
7    Cl    1.760000  1  119.886527  1  179.999999  1    2    1    3
8     H    1.070000  1  109.471203  1  179.999776  1    1    2    3
9     H    1.070000  1  109.471251  1   59.999791  1    1    2    3
10    H    1.070000  1  109.471218  1  299.999777  1    1    2    3
11
```

**Note:** Data formats often have a strictly defined syntax and can often only be processed if that syntax is followed to the letter!

Engel T. (2018) *Computer Processing of Chemical Structure Information* in Engel, T., Gasteiger, J., *Cheminformatics* (1st ed., pp. 43-119), Wiley-VCH.

# Acitivity: Data structures

1) Demo: Review different structure formats: .pdb, .mol, .mol2, .gjf

2) From a provided computational file (yne.gjf), use the connection data to derive the 2D structure of the molecule. What other data do you find in the file?

3) Discuss how you would be able to extract bond length for a specific atom pair from xyz-files.

To visualise 3D structures (and to build), you need software that can handle the structure formats, e.g.

- Chem3D
- Avogadro (https://avogadro.cc/)
- Jmol (https://jmol.sourceforge.net/)
- Pymol (https://pymol.org/)

Important principles:
Principles behind it yes, exact datastructure no
(ie principle question on connectivity matrices)

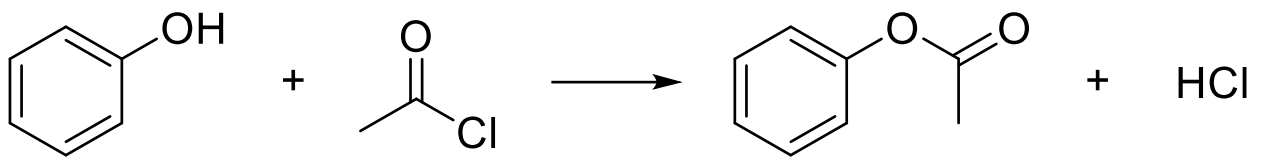
# Encoding chemical reactions

Reaction = mapping between two molecular states / a transformation in chemical space.

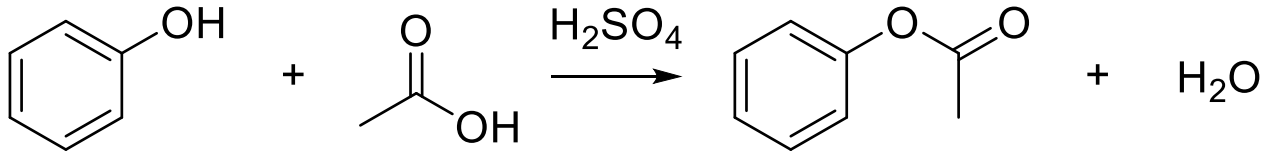
Reaction representations are important for **reaction prediction**, **classification**, and **retrosynthetic analysis**.

Representations:

1) Textual by components (columns)
2) Single string:
   - Reaction SMILES / SMARTS
   - RInChI (Layered string) / RInChIKey
3) Structured (e.g. arrays):
   - CML (similar to XML)
   - ORD Schema
4) Fingerprints SMILES will do the heavy lifting
5) Graphs




```
"c1ccccc1cO.CC(=O)Cl >> c1ccccc1OC(=O)C.Cl"
```




```
"c1ccccc1cO.CC(=O)O > OS(=O)(=O)O > c1ccccc1OC(=O)C.O"
```


Sacher O., Gasteiger J. (2018) *Representation of Chemical Reactions* in Engel, T., Gasteiger, J., *Cheminformatics* (1st ed., pp. 121-154), Wiley-VCH.
https://greglandrum.github.io/rdkit-blog/posts/2025-06-12-using-reaction-info.html
https://deepwiki.com/rdkit/rdkit-tutorials/2.3-smarts-reactions
https://www.daylight.com/meetings/summerschool01/course/basics/smirks.html

# Acitivity: Reaction encoding

1) Demo: How to work with a Reaction SMARTS in rdkit

2) Come up with SMARTS for these transformations:
   - Ester hydrolysis: C(=O)OR → C(=O)OH
   - Halogen exchange: Cl → Br
   - Bonus: Esterification R1OH + R2COOH → R2COOR1

3) What could these Reaction SMARTS be used for?

4) Continue with the SMARTS exercises

SMARTs content key content